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LETTERS

Measurement of the Instability Rate of a Far-from-Equilibrium Steady State at an Infinite Period Bifurcation

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We present the principles of a method to measure experimentally the instability rate of a steady state under far-from-equilibrium conditions. The method makes use of the period lengthening of a limit cycle occurring at a saddle-loop infinite period bifurcation. We illustrte the method by two examples of this phenomenon recently observed in oscillating chemical reactions.

When an autocatalytic chemical reaction is run far from equilibrium, for instance in a continuous-flow stirred tank reactor (CSTR), oscillations are often observed with a period which can be varied by changing one of the control parameters of the system, either the input concentration of one of the reactants or the flow rate. A bifurcation may then be generated where the period of the oscillation becomes arbitrarily large at a critical parameter value because the limit cycle merges with an unstable steady state[1,2] (see Figure 1). This so-called saddle-loop infinite period bifurcation has been the object of several recent publications.[3-5] However, the possibility of measuring the rate of instability of the associated steady state does not seem to have been reported. It is our purpose in this Letter to point out this important result[12] and to illustrate it with two experimental examples.

We present here a general theory which is model independent.[12-14] We assume that the particular far-from-equilibrium system is controlled by a parameter μ varying around μ_H, which is the critical value where the infinite period bifurcation occurs. The system also possesses an unstable steady state we choose as the origin of coordinate axes. From the general theory of dynamical systems, we know that the time behavior in its vicinity will be governed by its linear stability complex eigenvalues:[15,16]

$$..., \lambda'_s \pm i\omega'_s, \lambda_s \pm i\omega_s, \lambda_u \pm i\omega_u, \lambda'_u \pm i\omega'_u, ...$$

where

$$... \leq \lambda'_s \leq \lambda_s < 0 < \lambda_u \leq \lambda'_u \leq ...$$

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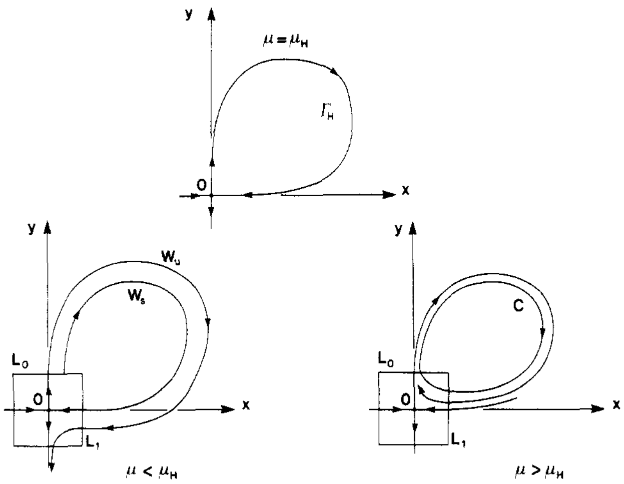

Figure 1. Schematic representation of a saddle-loop bifurcation in a two-dimensional phase space. The saddle point (0) is enclosed in a box with sides L_0 and L_1. A stable (W_s) and an unstable (W_u) manifold are associated with the saddle. Before the bifurcation ($\mu < \mu_H$), the vicinity of the saddle is unstable and trajectories escape to another attractor. At the critical configuration ($\mu = \mu_H$), a loop or homoclinic orbit (Γ_H) is formed which is a trajectory connecting the saddle point in both the past and the future. Beyond the bifurcation ($\mu > \mu_H$), a limit cycle (C) is generated.

These eigenvalues may depend slowly on the control parameter μ but we assume here that the real parts are not vanishing so that they cause no bifurcation. Besides this condition, we shall focus here on the case of interest for us, where

$$\omega_s = \omega_u = 0, \quad \text{and} \quad \lambda'_s < \lambda_s,\ \lambda_u < \lambda'_u$$

We shall make comments on the other cases at the end of this Letter.

The dynamics is known to be always dominated by the eigenvalues with the smallest real part in absolute value.[15,16] We shall thus simplify the argument and consider only the eigenvalues λ_s and λ_u and their associated eigendirections that we take as the axes x and y respectively of the coordinate system. The unstable steady state is then enclosed in a square box of side $2l$ where the kinetics is thus governed by the following differential equations:[12-16]

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \lambda_s x, \quad \frac{\mathrm{d}y}{\mathrm{d}t} = \lambda_u y \tag{1}$$

Any trajectory starting inside the box will soon escape along the unstable direction crossing one side of the box that we assume to be L_0 with $y = l$. The trajectory will then evolve in the phase space. After a while, the trajectory returns to the vicinity of the unstable steady state. It enters the box along the stable direction, crossing another side of the box we assume to be L_1 with $x = l$. The scenario repeats itself if the trajectory does not escape from the described part of the phase space. The flow can then be reduced into two mappings.

The first one is the mapping inside the box from L_1 to L_0. If y_1 is the y coordinate of a point on L_1 and if x_0 is the x coordinate of its images on L_0, the integration of the eqs 1 gives

$$M_{\mathrm{in}}:\quad L_1 \rightarrow L_0 \qquad \begin{cases} x_0 = l\left(\dfrac{y_1}{l}\right)^{|\lambda_s/\lambda_u|} \\ \tau_{\mathrm{in}} = \dfrac{1}{\lambda_u}\ln\dfrac{l}{y_1} \end{cases}$$

where τ_{in} is the time spent in the box.

The second mapping is generated by the flow outside the box between the sides L_0 and L_1. Near the infinite period bifurcation,

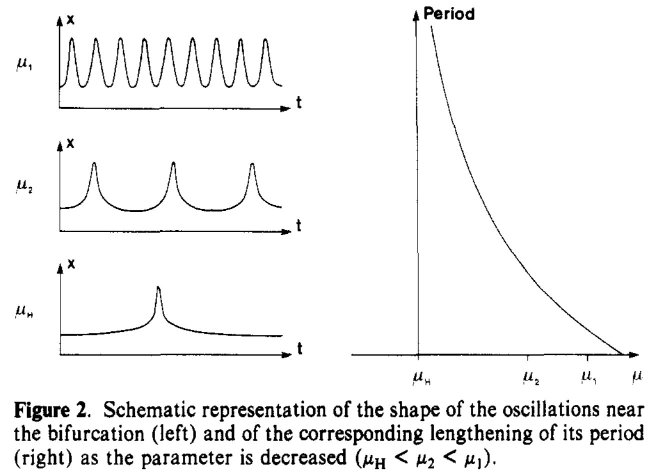

Figure 2. Schematic representation of the shape of the oscillations near the bifurcation (left) and of the corresponding lengthening of its period (right) as the parameter is decreased ($\mu_H < \mu_2 < \mu_1$).

the trajectories closely approach the unstable steady state so that they leave the box at small values of the coordinate x_0. Hence we may neglect the nonlinear terms of the second mapping and assume the following quasi-linear form:

$$M_{\mathrm{out}}:\quad L_0 \rightarrow L_1 \qquad \begin{cases} y_1 = a + b\mu + cx_0 \\ \tau_{\mathrm{out}} = \theta \end{cases}$$

The positive quantity θ is the time spent outside the box during the reinjection. According to the previous discussion, we suppose that b, c, and θ are nonvanishing slowly varying functions of μ and x_0. Furthermore, we shall later show that this assumption is in agreement with the experimental observations. We now choose b positive for definiteness. The phase space geometry of the system is represented schematically in Figure 1. For the critical value of the control parameter, namely,

$$\mu = \mu_H \simeq -a/b$$

the point $x_0 = 0$ of L_0 is mapped on $y_1 = 0$ of L_1 so that the corresponding trajectory forms a so-called loop or homoclinic orbit which is at the origin of the infinite period bifurcation.[12-16]

Combining M_{in} and M_{out}, we obtain the mapping from L_1 onto itself as

$$M_{\mathrm{out}} \text{ after } M_{\mathrm{in}}:\qquad y'_1 \simeq b(\mu - \mu_H) + cl\left(\frac{y_1}{l}\right)^{|\lambda_s/\lambda_u|} \tag{2}$$

The dynamical system possesses a periodic orbit if a solution exists of the equation formed by eq 2 with $y'_1 = y_1$. There are two cases: either (I) $|\lambda_s| < |\lambda_u|$ or (II) $|\lambda_s| > |\lambda_u|$. In the first case, there exists a periodic orbit but it is unstable. However, in the second case, the left-hand member of eq 2 is larger than the last term for small $y_1 = y'_1$. As a consequence, the periodic orbit crosses the side L_1 around $y_1 \simeq b(\mu - \mu_H)$ for $\mu > \mu_H$, but disappears when $\mu < \mu_H$. This periodic orbit is now stable and transforms into the homoclinic orbit as μ goes to the critical value μ_H while its period lengthens approximately like

$$T = \tau_{\mathrm{in}} + \tau_{\mathrm{out}} \simeq \frac{1}{\lambda_u}\ln\frac{l}{b(\mu - \mu_H)} + \theta$$

Close to the bifurcation, τ_{in} is greater and varies more rapidly than τ_{out}, whereupon

$$\mu - \mu_H \simeq Ce^{-\lambda_u T} \tag{3}$$

which proves our statement.[12] Accordingly, the period grows logarithmically as the bifurcation is approached. This behavior is represented schematically in Figure 2 with the corresponding shape of the oscillations. We observe that the instability eigenvalue λ_u appears as a rate constant in eq 3 so that we are able to measure it from the plot of Figure 2 if the saddle-loop bifurcation is applicable.

We illustrate this procedure with two examples.

The first one concerns the chlorite–thiosulfate–iodide oscillating reaction observed by Maselko and Epstein.[3] These authors re-

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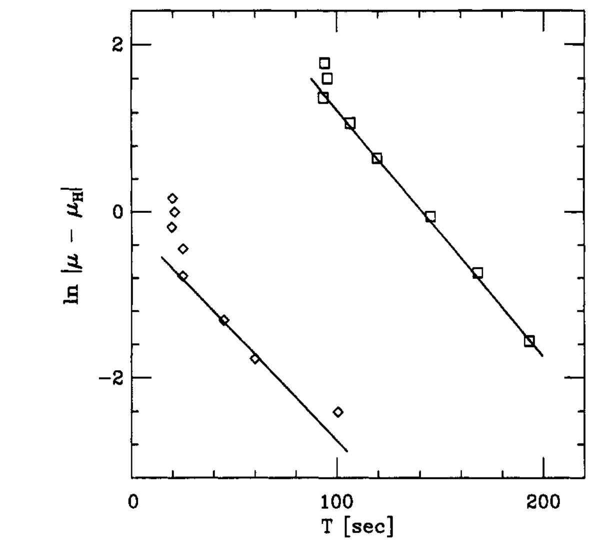

Figure 3. Plot of ln $|\mu - \mu_H|$ versus the period T of the oscillations for the data of the experiments by Maselko and Epstein (ref 3) on the chlorite–thiosulfate–iodide reaction (diamonds); and by Maselko, Alamgir, and Epstein (ref 4) on the bromate–iodide reaction (squares). The instability rates given in the text were inferred from the drawn solid lines. The logarithmic scale along the parameter axis contributes to the clustering of data points at high values of $|\mu - \mu_H|$.

ported a saddle-loop bifurcation in a CSTR of 27.5 mL at 25 °C and pH 4 for input concentrations $[S_2O_3^{2-}]_0 = 6 \times 10^{-3}$ M and $[I^-]_0 = 6.25 \times 10^{-2}$ M and a reciprocal residence time $k_0 = 5 \times 10^{-3}$ s^{-1}. The homoclinic orbit appears at $[ClO_2^-]_{0\,H} = 2.9 \times 10^{-2}$ M. We plotted these experimental results as $\ln\{10^2([ClO_2^-]_0 - [ClO_2^-]_{0\,H})\}$ versus the period T of the oscillation in Figure 3 (diamonds). From the long-period behavior, we extract the instability rate of the steady state to be $\lambda_u = (39\ \mathrm{s})^{-1}$.

The second example concerns the bromate–iodide oscillating reaction observed by Maselko, Alamgir, and Epstein[4] in a CSTR at 25 °C with input concentrations $[BrO_3^-]_0 = 4.375 \times 10^{-3}$ M, $[I^-]_0 = 5 \times 10^{-3}$ M, and $[H_2SO_4]_0 = 1.5$ M. The control parameter is the flow rate k_0 and the homoclinic orbit occurs at $k_{0\,H} = 22 \times 10^{-4}$ s^{-1}. In Figure 3 (squares), we plotted $\ln\{10^4(k_{0\,H} - k_0)\}$ versus the period T of the oscillations and we inferred the instability rate to be $\lambda_u = (34\ \mathrm{s})^{-1}$ in this system.

In Figure 3, the straight lines do not follow the experimental points when the period is small because the approximate eq 3 describes only the vicinity of the infinite period bifurcation. Far from the bifurcation, the unstable steady state does not perturb anymore the limit cycle so that the period is then slowly varying as observed in Figure 3. This fact supports our previous assumption that the mapping M_{out} is quasi-linear in these particular experimental conditions.

We think that the above-described method can be extended to other systems because the saddle-loop bifurcation has already been observed not only in several models of far-from-equilibrium chemical reactions[11,17] as well as in the theory of the isothermal[14,18] and nonisothermal[2,19] CSTR, but also in experiments such as the carbon monoxide–oxygen thermocombustion[20] besides the two afore-cited examples.

Some comments are now in order about the other cases with complex eigenvalues that we disregarded in our preceding analysis. A single periodic orbit is also generated via the infinite period bifurcation in the cases: (III) $\omega_s = 0$, $\omega_u \neq 0$, $|\lambda_s| < |\lambda_u|$; and (IV) $\omega_s \neq 0$, $\omega_u = 0$, $|\lambda_s| > |\lambda_u|$. The limit cycle is unstable in the case III but stable in the case IV where its period behaves then according to eq (3). However, chaotic behaviors with related period-adding bifurcation sequences are generated in the other cases: (V) $\omega_s = 0$, $\omega_u \neq 0$, $|\lambda_s| > |\lambda_u|$; (VI) $\omega_s \neq 0$, $\omega_u = 0$, $|\lambda_s| < |\lambda_u|$; and (VII) ω_s, $\omega_u \neq 0$.[13,21-23] Shil'nikov's chaos (case V) was observed in the Belousov–Zhabotinskii reaction[24] as well as in copper electrodissolution.[25] In these systems, the properties of the unstable steady state can be measured by statistical analysis of the time series.

We believe that the conclusions of the present paper are of special importance for the development of chemical kinetics in far-from-equilibrium conditions. When the conditions are satisfied for a saddle-loop bifurcation to occur, for instance in a CSTR, the method presented in this Letter allows us to measure experimentally the instability rate of the steady state. The experimental system could then be compared quantitatively with a kinetic model around this unstable steady state in order to fix the rate constants of the model. We find this method promising because linear stability of steady states can be predicted more easily than, for instance, the period of a limit cycle or even the location of a bifurcation.

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